

Lab: Brain Mapping -- Where Your Mental Models Live

Objective

Investigate how different **brain regions** contribute to the generative model, and explore how neuroscience reveals the biological basis of cognition as internal model-building.

Prerequisites

- Completed Biology modules on Systems, Agents, and Perception
- Read the module on Brain Regions and Mental Models

Part 1: Brain Region Function Map

Goal: Map key brain regions to their roles in the generative model.

Research each brain region and describe its contribution to cognition:

Brain Region	Location	What It Does	Role in Generative Model
Prefrontal cortex			High-level predictions, planning
Hippocampus			
Amygdala			
Cerebellum			
Thalamus			
Basal ganglia			

Write your response here

Part 2: The Hierarchical Brain

Goal: Understand how the brain builds models at multiple levels simultaneously.

The brain processes information hierarchically -- from simple features to complex concepts. Map the hierarchy for recognizing a friend's face:

1. **V1 (primary visual cortex):** What features does it extract?
2. **V2/V4:** What patterns does it assemble?
3. **Fusiform face area:** What does it recognize?
4. **Prefrontal cortex:** What context and memory does it integrate?
5. **Hippocampus:** What associations and memories does it retrieve?

How does this hierarchy relate to the Active Inference idea that the generative model has multiple layers, with each layer predicting the layer below?

Write your response here

Part 3: Neurotransmitters as Precision Signals

Goal: Connect brain chemistry to Active Inference precision-weighting.

Research the following neurotransmitters and their roles:

Neurotransmitter	What It Does	Active Inference Interpretation
Dopamine		Precision of predictions / expected reward
Serotonin		
Norepinephrine		
Acetylcholine		

How might an imbalance in these neurotransmitters change the brain's generative model?

Write your response here

Part 4: Cognitive Disorders as Model Failures

Goal: Analyze a cognitive condition as a failure of the generative model.

Choose one: ADHD, depression, schizophrenia, or OCD. Research and answer:

1. What aspect of the generative model is disrupted?
2. What prediction errors does the person experience?
3. How does the condition affect the precision-weighting of beliefs vs. sensory data?
4. How do treatments (medication, therapy) attempt to correct the model?

Write your response here

Summary Table

Concept	Brain Structure	Active Inference Role
Mental models	Prefrontal cortex + hippocampus	Generative model storage and updating
Hierarchical processing	Cortical hierarchy	Multi-level prediction
Precision-weighting	Neuromodulatory systems	Adjusting confidence in predictions vs. data
Cognitive disorders	Various	Disrupted generative model or precision